

Table 1: Examples from the collected benchmark dataset.

ID	Formula	Structure	Description	Band gap (eV)	Volume ($\text{\AA}^3/\text{cell}$)	Is-gap-direct?
mp-22851	NaCl		NaCl is Tetraauricupride structured and crystallizes in the cubic $P\bar{m}3m$ space group. Na^{1+} is bonded in a body-centered cubic geometry to eight equivalent Cl^{1-} atoms. All Na-Cl bond lengths are 3.03 $\text{\AA}$. Cl^{1-} is bonded in a body-centered cubic geometry to eight equivalent Na^{1+} atoms.	3.97	42.96	No
mp-30274	AcBrO		AcOBr is Matlockite structured and crystallizes in the tetragonal $P4/nmm$ space group. Ac^{3+} is bonded in a 4-coordinate geometry to four equivalent O^{2-} and five equivalent Br^{1-} atoms. All Ac-O bond lengths are 2.49 $\text{\AA}$. There are four shorter (3.40 $\text{\AA}$) and one longer (3.54 $\text{\AA}$) Ac-Br bond length. O^{2-} is bonded to four equivalent Ac^{3+} atoms to form a mixture of corner and edge-sharing OAc_4 tetrahedra. Br^{1-} is bonded in a 5-coordinate geometry to five equivalent Ac^{3+} atoms.	4.24	140.14	Yes
mp-570693	$\text{SbSe}_3\text{N}_2\text{Cl}_7$		$\text{SbCl}_6\text{N}_2\text{Se}_3\text{Cl}$ is Indium-derived structured and crystallizes in the triclinic $P1$ space group. The structure is zero-dimensional and consists of two $\text{N}_2\text{Se}_3\text{Cl}$ clusters and two SbCl_6 clusters. In one of the $\text{N}_2\text{Se}_3\text{Cl}$ clusters, N^{2+} is bonded in a bent 120 degrees geometry to one Se^{2+} and one Se^{3+} atom. The N-Se bond length is 1.75 $\text{\AA}$. The N-Se bond length is 1.75 $\text{\AA}$. N^{3+} is bonded in a bent 120 degrees geometry to one Se^{2+} and one Se^{3+} atom. The N-Se bond length is 1.74 $\text{\AA}$...	1.70	787.76	No

The data contains 144,931 crystals which we split into 125,098 crystals for training, 9,945 as a validation set, and 9,888 as a test set. For each crystal, we collect its ID, structural information, band gap, volume, and whether its band gap is direct or indirect. We extracted the crystal text descriptions using Robocrystallographer (Ganose and Jain, 2019) (see Table 1 for examples).

3.2 LLM-Prop Architecture

3.2.1 T5

T5 (Raffel et al., 2020) was the first unified language model that converts all text-based language problems into a text-to-text problem to perform both generative and predictive tasks. This framework is important for adapting and finetuning T5 on many tasks, and enables efficient multitask finetuning. Raffel et al. (2020) carefully compared different architectures, pretraining objectives, unlabeled datasets, transfer approaches, and more on dozens of natural language tasks and then combined the best-performing approaches in each comparison to pretrain T5. For instance, while MatBERT, a BERT-based model, was pretrained using a masked language model (MLM) objective (Devlin et al., 2018), T5 uses a span-masking objective (Joshi et al., 2020) as it was shown to outperform MLM objective in terms of predictive power and speed. These considerations motivate our choice to use T5 as our main pretrained model.

3.2.2 Adapting T5 for Predictive Tasks

Transforming each task to text-to-text format requires T5 to use its decoder part to generate the output. While the decoder is necessary for generative and many downstream tasks, it adds unnecessary memory, time complexity, and work overhead when adapted for predictive tasks. Furthermore, Raffel et al. (2020) argued that